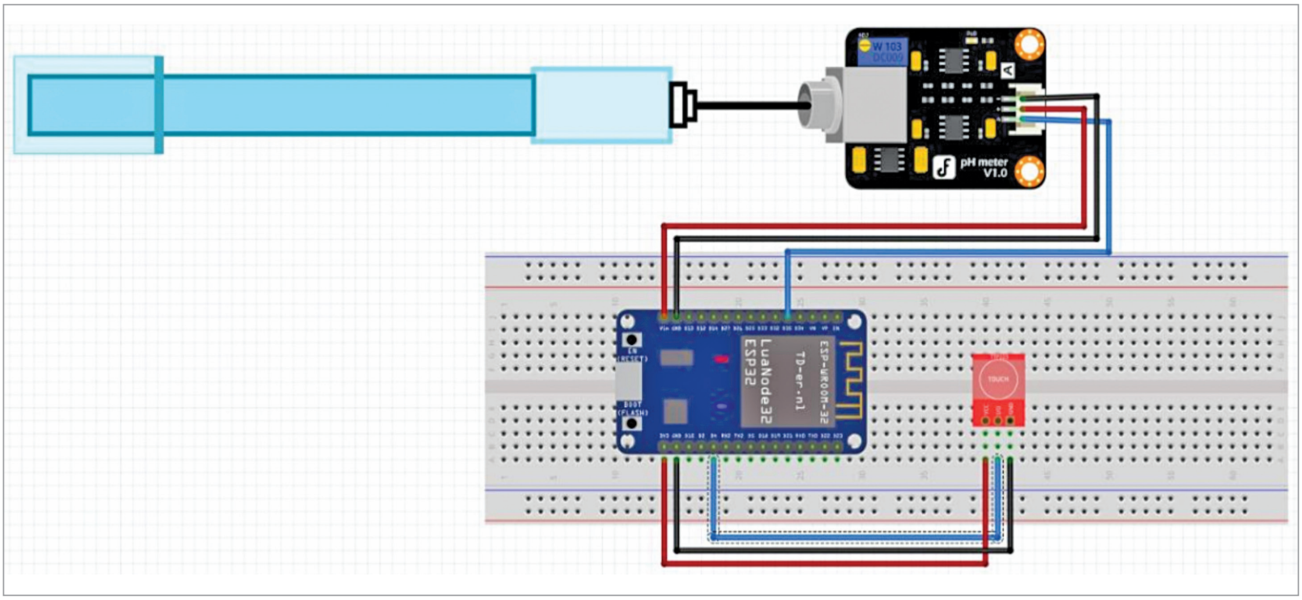

Fig. 7: Wiring of the proposed setup on a breadboard

memory is powered down. Only one RTC timer on the slow clock and specific RTC GPIOs are active. The RTC timer or RTC GPIOs can wake up the chip from hibernation mode.

The ESP32 deep sleep mode is utilised in the Arduino Cloud. In the current setup, the ESP32 microcontroller is initialised into deep sleep mode and awakened as needed. This can be achieved in three ways: timer wake up, touch wake up, and external wake up.

A comparison of the five different modes is provided in Table 1 from the ESP32 datasheet. The pinout is represented in Fig. 4, and Table 1 displays the various power modes and power consumption of the ESP32 chip.

Timer wake up, touch wake up, and external wake up

Every setup has unique requirements. In this setup, the pH of the water circulated in the hydroponics setup is monitored using the E201 BNC. The ESP32 is maintained in deep sleep mode to conserve power consumption. The microcontroller wakes up using the external mode, which involves an attached touch sensor. Upon touch, the pH reading is taken, and the value is displayed on the serial monitor.

TABLE 2
BILL OF MATERIALS

Components	Amount	Description
ESP32 (MOD2)	1	For programming
pH sensor (MOD3)	1	To sense the pH value
Touch sensor (MOD1)	1	To initiate an external input for taking readings
Jumper wires (male to female)	3	To connect the sensor

During deep sleep, some of the ESP32 pins can be utilised by the ULP coprocessor, namely, the RTC_GPIO pins and the touch pins. The ESP32 datasheet's pinout identifies the RTC_GPIO pins, which are highlighted in a green rectangular box.

External wakeup

External wakeup is another method of waking up the ESP32 board from deep sleep mode, where a change in a GPIO pin's state triggers the awakening. The wakeup source must be configured before setting the ESP32 board into deep sleep mode. Two types of external wakeups can be set up: ext0 and ext1. Ext0 involves configuring one GPIO pin as an external wakeup, while ext1 is used for multiple GPIO pins. It is important to note that only RTC GPIO pins can be used as a wakeup source for external wakeup.

TABLE 3
CONNECTIONS BETWEEN MOD1, MOD2, AND MOD3

MOD3 (pH Sensor)	MOD2 (ESP32)
Voltage supply (5V)	Vcc (5V)
Output of sensor (A)	G35
GND	GND
MOD1 (Touch Sensor)	**MOD2 (ESP32)**
Voltage supply (Vcc)	Vcc (5V)
Output of sensor (SIG)	G4
GND	GND

To enable the ext0 wakeup source, use the following API:

```
esp_sleep_enable_ext0 wakeup()
```

The RTC IO module is responsible for triggering the ext0 wakeup. As the RTC IO module is part of the RTC peripherals power domain, the RTC peripherals must be powered